TECHKRITI'27

ROBOGAMES: IARC PROBLEM STATEMENT

> *"Artificial intelligence learns from me, but I'm still learning what it truly is."*
> ***— Dave Waters***

Introduction

This event challenges participants to design and build an autonomous robot from scratch. The robot is judged based on its ability to perform specific traversal tasks, demanding precision in fabrication and intelligent use of sensors and coding. As one of the flagship events of Techkriti, IARC continues to push the boundaries of collegiate robotics.

General Rules

- Each team can have a maximum of 6 participants.
- A team may consist of students from different colleges.
- Certificates of Excellence will be awarded to the top three teams.
- Bots should not be disassembled until the results are declared.
- When a team is called for a match, they must report within five minutes.
- The actual arena may differ slightly from reference diagrams, but the core tasks and dimensions will remain the same.
- The organizers reserve the right to change the rules as they deem fit. Changes will be highlighted on the official website.
- The judge's decision is final and binding in all cases.

Robot Specifications & Controls

- **Dimensions:** During the whole event, the bot must fit within a square volume of 250 × 250 × 250 (mm) (length × width × height).
- **Autonomy:** The robot must be stable and able to move on its own. You have to let go of the bot at the beginning of the trial; no human intervention is allowed thereafter. If human intervention is necessary, a penalty will be

Problem Statement & Node Decoding

You shall place your bot at the 'START' zone in any direction to begin line following. Along the path, there will be a total of 3 specific nodes that the bot must decode using its sensors.

- The node is a 2×2 matrix that your bot must sense and store.
- Your bot will cross a node by first encountering the C & D cells, followed by the A & B cells.
- You must convert that matrix into a 4-bit binary number. If the number in binary form is ABCD, A, B, C, & D represent the bits detected in the following order:

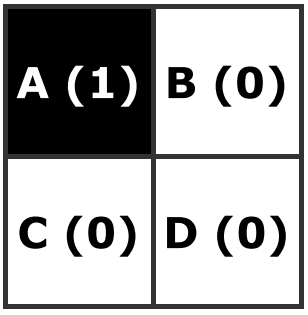


Fig: Top View of Node Matrix Example

- For example, in the above figure, the value read will be **1000** (A=1, B=0, C=0, D=0), where black is 1 and white is 0.
- **Data Display:** Your bot must display the binary value of each node in real-time. This can be done via an onboard LCD screen, or transmitted via wireless communication to a mobile/laptop screen.

Navigating the Path

Dead Ends (Obstacle Detection): There will be several dead ends in the maze path denoted by a cubic wooden obstacle. The robot must detect this obstacle and reverse its course without touching it. Touching the block will result in a penalty.

Task Flow: Speed Zones and Angles

- **Node A to Node B (Speed Modulation):** After reaching point A, the robot will encounter a node which it must decode into a binary value, and then convert to a decimal number, *x*. The bot must then travel to point B following the black line, taking **at least** (x × 2) seconds. If the bot completes the distance faster than this time threshold, a penalty is deducted.

- **Node B:** After reaching point B, the bot will encounter a node with value 0000. This signals that the speed limit zone is over. The bot resumes standard line following to reach point C.

- **Node C (Lane Selection):** After reaching C, the bot will detect a final node and decode it into a decimal number, *y*. The correct exit lane to the goal is located at an angle of **y × 15 degrees** from the entry lane. The bot must follow a circular path and turn into this correct lane (which may be clockwise or anticlockwise).

- **The Finish:** Following the decoded angle lane will lead the bot to the Finish area. Upon stopping, it must display the message "FINISH" on its LCD or connected screen.

Arena Specifications

- The width of the black line is 3 cm.

- The size of each node is 6 cm × 6 cm, and each individual square cell in the node has dimensions of 3 cm × 3 cm.

- The size of dead-end cubic obstacles is 150 mm × 150 mm × 150 mm.

- The distance between nodes B and C can vary from 50 cm to 150 cm.

- All checkpoints (including start and end points) are solid circular nodes of 30 cm dimension and are black in color.

- The radius of the circular path following Node C is 200 mm.

Abstract Submission

Once registered, all teams must submit a PDF document of their abstract to qualify for the event.

- The abstract must be submitted only once from the team leader's dashboard.

- The file size must be less than 3MB in .pdf format.

- The team leader must confirm the Tech ID of all registered members before submission.

Updated T27 Scoring System

To reward consistent problem-solving, base points are awarded for checkpoint reach, with time counting as a positive bonus rather than a deductive base.

Base Zone Points:

- Checkpoint A Reached: +500 points
- Checkpoint B Reached: +500 points
- Checkpoint C Reached: +500 points
- Finish Reached: +1000 points

Time Bonuses (Time taken in seconds):

- Start to A (t_1): Bonus = $(200 - t_1)$. If $t_1 > 200$, Bonus = 0.
- B to C (t_3): Bonus = $(200 - t_3)$. If $t_3 > 200$, Bonus = 0.
- C to Finish (t_4): Bonus = $(300 - t_4)$. If $t_4 > 300$, Bonus = 0.

Penalties:

- **Human Intervention:** -50 points per instance.
- **Obstacle Touch:** -50 points per instance (Dead ends).
- **Speed Zone Violation:** -150 flat penalty if time taken from A to B is LESS than $(x \times 2)$ seconds.
- **Wrong Exit Angle:** -100 points for selecting the incorrect lane at Node C.

Final Score = Total Base Points + Total Time Bonuses − Total Penalties

Contact Information

For any queries regarding the technical specifications or event structure, please reach out to the managing team or mail us at robogames@techkriti.org.

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